BKM-Aware MAAT Diagnostics for Navier–Stokes

A Phenomenological Structural-Action Test on Taylor–Green Vortices

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Abstract

This note develops a phenomenological MAAT v1.6 diagnostic for the three-dimensional incompressible Navier–Stokes equations. It is motivated by the symbolic structural-action expression

$$\mathrm{ToE}_{\mathrm{MAAT}} = \int \frac{(H + B + S + V + R)\, Z_{\mathrm{MAAT}}\, \chi_{\mathrm{cascade}}}{\Delta E + \Delta Q + D_0} \mathrm{d}t,$$

where χ_{cascade} is a cascade-resistance factor built from spectral-tail and vortex-stretching pressure, and $D_0 = 11$ is treated as a fixed dimensionless baseline penalty.

The construction is not a proof of Navier–Stokes regularity. Instead, it asks whether MAAT structural coordinates can act as time-dependent navigation and warning coordinates along a fluid trajectory. This is not evidence for a fundamental theory; it is a test of whether a structural-action expression can be converted into a measurable fluid diagnostic. We implement a pseudo-spectral incompressible solver on a periodic box and monitor Taylor–Green vortex evolutions in two regimes: moderate viscosity and lower-viscosity/high-stress flow. The diagnostic tracks equation consistency, incompressibility, energy balance, bounded activity, low-mode spectral coherence, robustness closure, vorticity-stretching pressure, and a Beale–Kato–Majda integral proxy.

The main result is qualitative but useful. The MAAT warning is not simply a proxy for the vorticity infinity norm. In both tested trajectories it correlates much more strongly with vorticity-stretching pressure: $\rho_{\mathrm{S}} = 0.9912$ in the moderate-viscosity run and $\rho_{\mathrm{S}} = 1.0000$ in the high-stress run. The integrated $\mathrm{ToE}_{\mathrm{MAAT}}$ action is larger for the more coherent moderate-viscosity trajectory (0.9567) than for the high-stress trajectory (0.6786). Thus the structural-action expression can be converted into a reproducible phenomenological diagnostic, but not into a theorem. This stretching sensitivity is the structural value added by the diagnostic: it points to a genuinely three-dimensional stress channel rather than merely renaming a standard vorticity monitor.

1 Purpose

The incompressible Navier–Stokes equations are

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \tag{1}$$

$$\nabla \cdot u = 0. \tag{2}$$

The Clay regularity problem asks whether smooth three-dimensional initial data can develop singularities in finite time. The present paper does not solve that problem. Its purpose is more modest: to build the strongest MAAT v1.6 diagnostic we can reasonably build at toy scale and ask what it actually measures. The intended value is not a larger vocabulary for known quantities. The intended value is to test whether the MAAT warning isolates structural risk channels, especially vortex stretching, that are not captured by a simple monotone reading of $\|\omega\|_\infty$.

MAAT v1.6 treats dynamics as search geometry. A fluid solution is therefore read as a trajectory

$$u_0 \to u_1 \to u_2 \to \cdots$$

through a high-dimensional space of velocity fields. Structural supports are not merely labels on a completed trajectory. They become navigation coordinates indicating whether the evolution remains in a coherent region or drifts toward structural stress.

2 Symbolic Functional

The symbolic starting point is

$$\mathrm{ToE}_{\mathrm{MAAT}} = \int \frac{(H + B + S + V + R)\, Z\, (\text{cascade resistance})}{\Delta E + \Delta Q + D_0} \mathrm{d}t.$$

Here D_0 is not assigned any microscopic interpretation. It is a fixed dimensionless baseline penalty that prevents the denominator from being dominated by small numerical defects in benign trajectories. For a numerical fluid diagnostic we operationalise it as

$$\boxed{\mathrm{ToE}_{\mathrm{MAAT}} = \int \frac{(H + B + S_{\mathrm{eff}} + V + R_{\mathrm{rob}})\, Z_{\mathrm{MAAT}}\, \chi_{\mathrm{cascade}}}{\Delta E + \Delta Q + D_0} \mathrm{d}t, \qquad D_0 = 11.} \tag{3}$$

This is a phenomenological score, not a fundamental action. The numerator rewards coherent structural support; the denominator penalises energy-balance defect, spectral-tail/numerical-quality defect, and the fixed baseline penalty.

3 Operational Supports

The supports are defined as follows:

- H: equation residual and incompressibility consistency.
- B: energy-balance consistency.
- S_{eff}: bounded activity pressure from palinstrophy.
- V: low-mode spectral coherence.
- R_{rob}: v1.2.1 robustness closure.
- Z_{MAAT}: structural partition factor.
- χ_{cascade}: cascade resistance from spectral-tail and vortex-stretching pressure.

The robustness closure is

$$R_{\mathrm{resp}} = (HBV)^{1/3}, \qquad R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}}, (HBS_{\mathrm{eff}}V)^{1/4}\Big). \tag{4}$$

The warning functional is

$$W_{\mathrm{MAAT}} = (1 - R_{\mathrm{rob}})\sqrt{A}\,(1 + \Theta_{\mathrm{tail}})\,(1 + c_s\Pi_{\mathrm{stretch}}), \tag{5}$$

where A is activity pressure, Θ_{tail} is spectral-tail pressure, and Π_{stretch} is vortex-stretching pressure.

4 BKM and Vortex Stretching

The Beale–Kato–Majda criterion states that loss of smoothness for the inviscid Euler equation is connected to divergence of

$$\int_0^T \|\omega(t)\|_\infty \,\mathrm{d}t, \qquad \omega = \nabla \times u.$$

For Navier–Stokes this is not used here as a theorem, but as a diagnostic risk channel. The more specifically three-dimensional term is vortex stretching,

$$(\omega \cdot \nabla)u.$$

A useful MAAT warning should not merely track vorticity magnitude. It should respond to structural channels that can transfer coherent motion into small-scale stress.

5 Numerical Benchmark

The experiment uses a pseudo-spectral incompressible solver on a periodic three-dimensional box with dealiasing and projection onto the divergence-free subspace. The initial condition is a Taylor–Green vortex. Two trajectories are tested:

Scenario	**Viscosity**	**Amplitude**
moderate viscosity	0.020	1.00
low viscosity / high stress	0.006	1.25

6 Results

Table 1: MAAT–Navier–Stokes diagnostic summary.

Quantity	**Moderate**	**High stress**
final energy	0.0894	0.1809
maximum $\|\omega\|_\infty$	1.9960	2.4991
final BKM proxy integral	3.6937	3.7750
minimum R_{rob}	0.9029	0.7578
maximum MAAT warning	0.1677	0.5692
final $\mathrm{ToE}_{\mathrm{MAAT}}$ action	0.9567	0.6786
final warning action	0.1052	0.2124
maximum spectral-tail fraction	0.00130	0.00372
Spearman warning vs. $\|\omega\|_\infty$	−0.7533	−0.4793
Spearman warning vs. vortex stretching	0.9912	1.0000
Spearman $\mathrm{ToE}_{\mathrm{MAAT}}$ integrand vs. R_{rob}	0.9411	1.0000

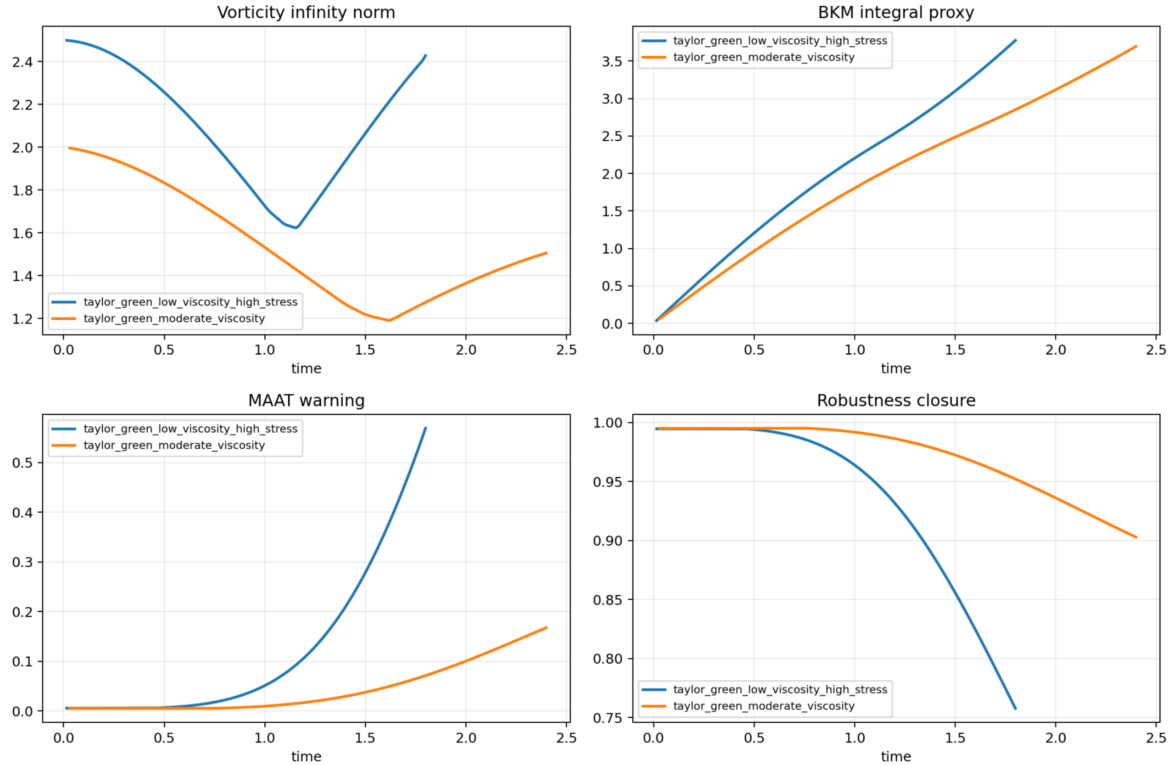

Figure 1: BKM-aware diagnostic time series. The high-stress trajectory develops larger warning and lower robustness than the moderate-viscosity trajectory.

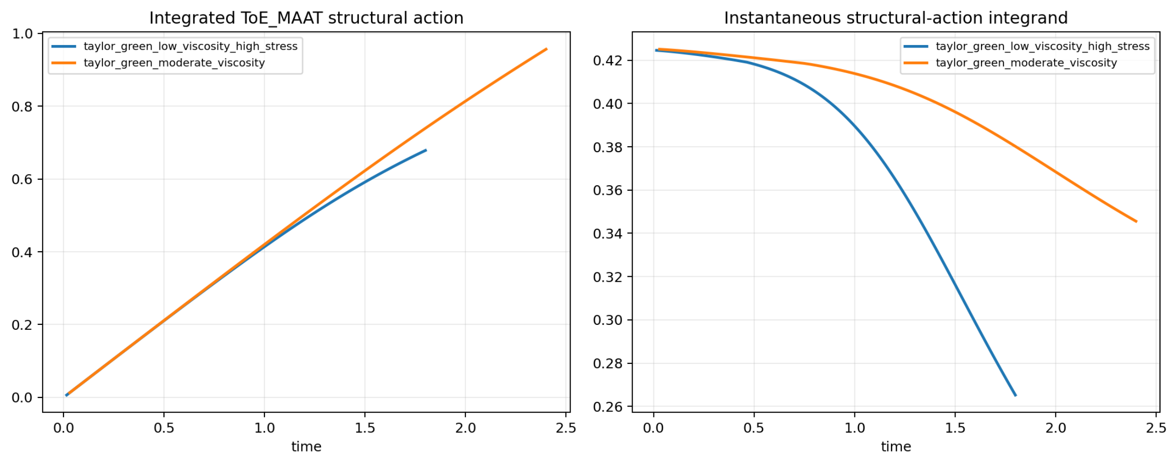

Figure 2: Integrated symbolic $\mathrm{ToE}_{\mathrm{MAAT}}$ action and instantaneous integrand. The more coherent moderate-viscosity trajectory accumulates larger structural action.

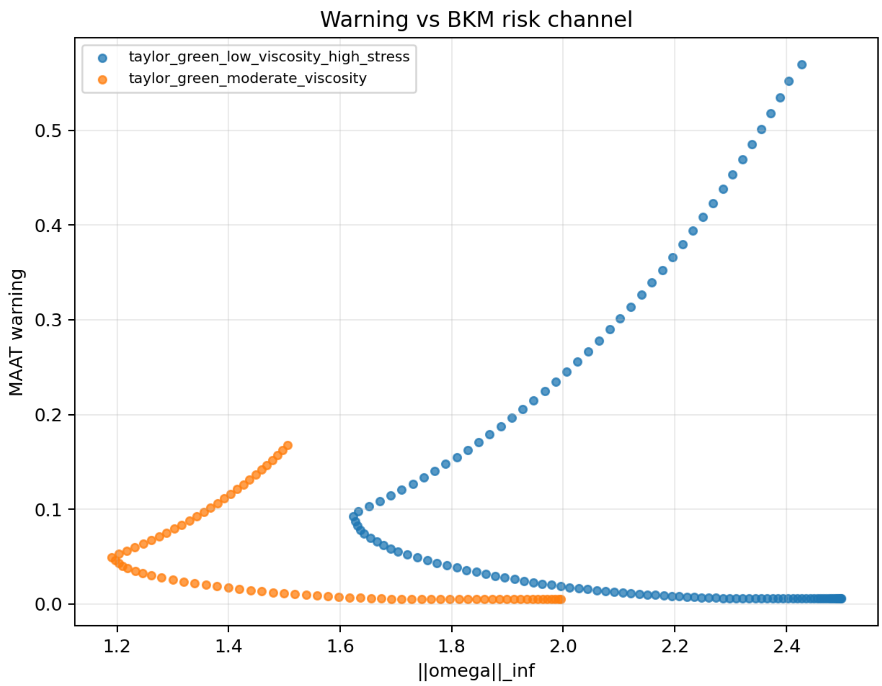

Figure 3: Warning versus vorticity infinity norm. The warning is not simply a monotone vorticity proxy; it is more closely tied to vortex-stretching pressure in this benchmark.

7 Interpretation

The most important result is negative in a useful way: W_{MAAT} does not simply track $\|\omega\|_\infty$. This matters because a naive blow-up diagnostic would merely duplicate the BKM quantity. Instead, the warning aligns with vortex stretching, the genuinely three-dimensional structural-risk channel. That is the main structural value added by the diagnostic. The observed correlations should not be interpreted as evidence for singularity prediction. They indicate only that the structural diagnostic appears sensitive to the dynamically relevant stretching channel in these controlled toy trajectories.

The symbolic $\mathrm{ToE}_{\mathrm{MAAT}}$ functional also behaves in the intended direction. It assigns higher accumulated structural action to the more coherent moderate-viscosity trajectory and lower action to the high-stress trajectory.

8 Limitations

- The simulation is low-resolution and intended only as a toy benchmark.
- Only two Taylor–Green trajectories are tested, so the correlations are not yet statistically broad evidence.
- Taylor–Green vortex flow is not a proof-level regularity test.
- The denominator term $D_0 = 11$ is a fixed numerical baseline penalty, not a derived physical constant.
- The cascade-resistance term is phenomenological and should be stress tested against alternative definitions.
- No theorem follows from the numerical correlations.

- A serious next step would require resolution studies, forced turbulence, adaptive grids, and public direct-numerical-simulation datasets.

9 Reproducibility

The experiment is stored in:

```
experiments/navier_stokes_maat_regularization/
```

Run:

```
cd experiments/navier_stokes_maat_regularization
python3 navier_stokes_maat_regularization.py
```

Outputs are written to:

```
outputs_phenomenological/
```

The public repository is:

https://github.com/Chris4081/structural-selection-principle

No external fluid dataset is redistributed. Velocity fields are generated locally from Taylor–Green initial conditions. CSV, JSON, and PNG files are derived reproducibility artifacts.

10 Conclusion

This phenomenological paper takes the symbolic MAAT expression seriously enough to operationalise it, but not so literally that it becomes an overclaim. The result is a BKM-aware structural diagnostic for three-dimensional Navier–Stokes trajectories.

The strongest finding is that the MAAT warning aligns with vortex stretching rather than merely duplicating vorticity magnitude. The integrated $\mathrm{ToE}_{\mathrm{MAAT}}$ action separates a more coherent trajectory from a high-stress trajectory in the expected direction. This is a meaningful next step for MAAT v1.6 as search geometry over evolving fields. It is not a solution of Navier–Stokes.

References

[1] C. L. M. H. Navier, "Memoire sur les lois du mouvement des fluides," *Memoires de l'Academie Royale des Sciences de l'Institut de France* **6**, 389–440 (1822).

[2] G. G. Stokes, "On the theories of the internal friction of fluids in motion," *Transactions of the Cambridge Philosophical Society* **8**, 287–305 (1845).

[3] J. T. Beale, T. Kato, and A. Majda, "Remarks on the breakdown of smooth solutions for the 3-D Euler equations," *Communications in Mathematical Physics* **94**, 61–66 (1984).

[4] P. Constantin and C. Foias, *Navier–Stokes Equations*, University of Chicago Press, Chicago (1988).

[5] M. E. Brachet, D. I. Meiron, S. A. Orszag, B. G. Nickel, R. H. Morf, and U. Frisch, "Small-scale structure of the Taylor–Green vortex," *Journal of Fluid Mechanics* **130**, 411–452 (1983).

[6] C. L. Fefferman, "Existence and Smoothness of the Navier–Stokes Equation," Clay Mathematics Institute problem description (2006).